

NATHANIEL W. BROOKS

SOFTWARE ENGINEER

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EXPERIENCE

GoPanda — Full Stack Engineer

NOV 2025 — PRESENT

- Ship production full-stack features across React/TypeScript web and React Native mobile clients, maintaining both builds in tandem on a Node/Express/Sequelize backend with PostgreSQL.
- Architect real-time features on WebSockets with Redis pub/sub for horizontal scaling, and offload heavy processing to Redis-backed job queues to keep API responses fast.
- Build and maintain bespoke internal admin tooling with live-updating operational views, providing real-time visibility into platform activity.
- Expand unit and containerized integration test coverage, and review teammates' pull requests to keep releases stable.

Fortinet Federal — Product Management & Cloud Intern

JUN — AUG 2025

- Owned one end-to-end project: architected and built a policy-compliant full-stack automation platform for firewall configuration and isolated network provisioning.
- Implemented infrastructure-as-code with Terraform, backed by FastAPI services and a Svelte frontend.
- Led system design and UI/UX within federal compliance frameworks including FedRAMP, ZTNA, SASE, CMMC, and NIST.

Kite Technology Group — Security & Operations Intern

MAY 2024 — MAY 2025

- Monitored client infrastructure in NOC operations, handling 250+ tickets a week and proactively addressing issues such as low disk space, UPS battery failures, and internet outages to keep clients online.
- Supported security and compliance audits through internal cybersecurity training campaign deployment and documentation.
- Maintained clients' Microsoft 365 environments using Azure, Intune, and Entra.

PROJECTS

Ledger — Cross-platform time tracking with shared workspaces, threaded comments, and markdown notes. Stateless Fastify/TypeScript API on PostgreSQL with Redis-backed job workers, one SvelteKit codebase shipping as PWA and native Tauri desktop app.

Spark — Real-time matchmaking with live user queues, session orchestration, and WebSocket messaging on a FastAPI and PostgreSQL backend with a React frontend.

Joust — Physics-based, pixel-styled jousting game in Three.js with custom GLSL/WebGL shaders, dynamic camera transitions, and real-time collision.

Caption Creators — Round-based multiplayer captioning game using AI-generated prompts and the YouTube API to serve videos, built with Next.js and MySQL.

GrubGuesser — SwiftUI game for guessing and comparing foods' nutritional values, built in MVVM with a FastAPI backend serving USDA FoodData Central data.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Java, Python, Swift, SQL
Frontend	React, React Native, Svelte 5 / SvelteKit, Next.js, TailwindCSS, Three.js, SwiftUI
Backend	Node.js, Express, Fastify, FastAPI, WebSockets, Redis queues
Data & Infra	PostgreSQL, MySQL, Redis, Sequelize, Drizzle, AWS, Docker, Terraform, Tauri

EDUCATION

Towson University — B.S. in Computer Science

3.82 GPA